# RESIAYU KINASIH PANGASTUTI

**Data Analyst | Business Intelligence | Oil & Gas Analytics**

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## PROFESSIONAL SUMMARY

Data Analyst and Business Intelligence professional with 7+ years in upstream oil & gas, specializing in Power BI, SQL, and Python for production, reservoir, and operational analytics. Turns complex field data into dashboards and decisions: prioritized 293 high-potential steamflood opportunities across 700+ patterns, cut a reporting cycle by 83%, and completed verification of a 48-well field data repository on target. Based in Indonesia (UTC+7) and open to full-time remote data analyst and business intelligence roles.

## CORE SKILLS

**Data Analytics:** SQL (Joins, CTEs, Window Functions, Subqueries, Aggregations), Data Cleaning, Data Verification and Validation, Statistical Analysis, Exploratory Data Analysis

**Business Intelligence:** Power BI (Data Modeling, Star Schema, DAX, Power Query/ETL, Interactive Dashboards, Drill-through, Row-Level Security), Spotfire, KPI Reporting

**Python & Machine Learning:** Python (pandas, scikit-learn, XGBoost), Classification, Class Imbalance Handling, Hyperparameter Tuning, Model Evaluation

**Data Management:** Data Quality, Data Integration, Data Repository Management, Operational Data Management

**Domain:** Production Analytics, Reservoir Analytics, Well Performance, Steamflood Surveillance, Petrel

**Languages:** Indonesian (Native); English (Intermediate)

## PROFESSIONAL EXPERIENCE

**PT Pertamina EP Cepu Zona 11 – Head Office** **Jun 2024 – Present**

*Data Analyst – Well Operation, Petrophysics and Data Management*

- Completed, with the team, the verification and completion of the well data repository for an entire field (47 wells) on target, consolidating data from multiple sources.
- Streamlined reservoir data collection and management for daily operations, improving accessibility for cross-functional analysis in a large-scale field environment.
- Designed a structured, dynamic data organization system that improved data accessibility and the efficiency of cross-functional team analysis.
- Managed critical data uploads to the UDIS ODR system, maintaining data integrity and availability for operational reporting and analysis.

**PT Pertamina Hulu Rokan Zona 2 & 3 – Duri Field** **Mar 2022 – Dec 2023**

*Petroleum Engineer – Asset Optimization Heavy Oil*

- Analyzed 700+ patterns through integrated reservoir-production surveillance and prioritized 293 high-potential steamflood opportunities for field optimization.
- Developed interactive Power BI dashboards that automated production monitoring and anomaly detection, helping operational teams identify production exceptions more efficiently.
- Monitored and evaluated steamflood injection and production performance to keep it aligned with reservoir management goals.
- Collaborated with cross-functional teams to design corrective action plans for underperforming patterns, driving field optimization.

**PT Pertamina Hulu Rokan Zona 2 & 3 – Minas Field** **Apr 2019 – Feb 2022**

*Petroleum Engineer Support – Analytical Support Team IEM*

*Concurrent assignment (Nov 2020 – Aug 2021): also worked in the Well Reliability Optimization team (next entry) to provide temporary coverage for retiring staff.*

- Digitized operator daily checklists, reducing the reporting cycle time by 83% (from 30 minutes to 5 minutes).
- Built interactive Power BI dashboards that automated production monitoring and anomaly detection for operational users.
- Delivered Power BI training to engineers and operations teams to support self-service analytics and knowledge transfer.

- Executed real-time production surveillance and exception management using Integrated Exception Management (IEM) systems.

**PT Pertamina Hulu Rokan Zona 2 & 3 – Duri Field** **Nov 2020 – Aug 2021**

*Petroleum Engineer Support – Well Reliability Optimization (Heavy Oil)*

*Held concurrently with the Minas Field role: worked across two teams to provide temporary coverage for retiring staff.*

- Optimized Sucker Rod Pump (SRP) performance through evaluation of dynamometer data and diagnostic analysis.
- Supported well service programs and pump replacements to restore offline or low-performing wells quickly.
- Initiated a best-practice verification protocol for dynamometer conditions, reducing operational risks and errors in well-status reporting.

**PT LAPI ITB – Bandung** **Aug 2018 – Mar 2019**

*Junior Reservoir Engineer*

- Ran reservoir simulations and data processing to evaluate reservoir behavior.
- Performed history matching to validate simulation models and improve the accuracy of reservoir characterization.
- Used Petrel for production forecasting to support long-term strategic planning.

**Institut Teknologi Bandung – Bandung** **Jan 2016 – May 2016**

*Assistant of Statistical Laboratory*

- Delivered briefings on data processing methods and practices to support statistical accuracy.
- Tutored second-year students in statistical software and data interpretation.

## PROJECTS

**Symptom-Based Illness Classification (Machine Learning)** **Personal project**

*Python, pandas, scikit-learn, XGBoost | github.com/resiayukp/symptom-classification-logistic-regression*

- Compared eight classification algorithms (KNN, Logistic Regression, SVC, Decision Tree, Random Forest, Gradient Boosting, XGBoost, MLP) on a synthetic, imbalanced dataset of 44,453 records and 20 symptom features, tuning each with cross-validated grid search.
- Applied class weighting to handle severe imbalance (COVID 5%, COLD 2% of records): COVID recall rose from 0.52 to 0.95 and COLD recall from 0.64 to 1.00 while accuracy stayed at 0.93; balanced accuracy rose from 0.77 to 0.96.
- Found that six models made identical predictions at the dataset's ~93% accuracy ceiling and selected Logistic Regression as the simplest, fastest, and most interpretable option (test accuracy 0.932, balanced accuracy 0.956, macro F1 0.812).
- Interpreted model coefficients to show which symptoms drive each prediction and documented limitations, including a precision of about 0.5 for COLD and COVID predictions.

**Volve Field Production Analytics Dashboard (Power BI)** **Personal project**

*Power BI, DAX, Power Query | github.com/resiayukp/volve-field-production-analytics-dashboard*

- Built a four-page interactive dashboard on Equinor's open Volve dataset covering oil, gas, and water production and water injection, with well-level and spatial analysis.
- Designed a star-schema data model and DAX measures for production trends, well performance, and operational KPIs.
- Implemented data transformation and validation workflows in Power Query.

## EDUCATION

**Institut Teknologi Bandung (ITB)** **2018**

*Bachelor of Science in Petroleum Engineering | GPA: 3.57 / 4.00*

## PUBLICATIONS

- Adli Muharam, C. and Pangastuti, R.K. (2023) 'Optimization of DSF remaining opportunities by integrated analysis of dynamic well data and reservoir data', Proceedings of the Indonesian Petroleum Association, 47th Annual Convention & Exhibition, Jakarta, Indonesia.

- Nainggolan, S.A.S., Wilantara, D., Mutiaranti, R. and Pangastuti, R.K. (2023) 'Artificial Intelligence and data analysis approach to deliver 5800 BOPD from cyclic steam stimulation job in Krakatau Field', Proceedings of the Indonesian Petroleum Association, 47th Annual Convention & Exhibition, Jakarta, Indonesia.

**TRAINING**

- Microsoft Learn: Data Analyst learning path (Power BI, Data Modeling, DAX)